

Saliency Maps and Input-Space Interpretability Adversarial Quiz

Question 1: Which ONE of the following principles ensures that two identical networks receive the same attributions?

- A. **Sensitivity** – Only knows whether a feature has an impact on the output
- B. **Completeness** - Focuses on the total overall amount of attribution
- C. **Implementation Invariance** - Attribution depends on the network's function and input, not the specific implementation details.
- D. **Input Invariance** – Confuses internal model structure with external input behaviour

Pitfall: Confusing model equivalence with input behaviour.

Question 2: Which ONE of the following best describes how SmoothGrad reduces the visual noise found in standard gradient-based sensitivity maps?

- A. **Averaging sensitivity maps of an image with added Gaussian noise** – It uses the formula $\hat{M}_c(x) = 1/n \sum_{i=1}^n M_c(x + N(0, \sigma^2))$ to smooth out sharp fluctuations.
- B. **Placing a Gaussian blur to the saliency map after calculating the raw gradient** – This is a common visual misconception.
- C. **Multiplying the gradient by the input image pixels** – Describes the "Input x Gradient" method.
- D. **Discarding all negative gradient values during back propagation** – Used in Guided Backprop, not SmoothGrad

Pitfall: Assuming smoothing = filtering instead of statistical averaging.

Question 3: Which ONE of the following is a characteristic of 'intrinsic interpretability'?

- A. **It "interrogates" complex, pre-trained models that aren't naturally easy to read.** – Intrinsic models are simple to understand just by looking at them (e.g., decision trees).
- B. **It works for deep neural networks.** - Post-hoc interpretability is more useful for complex models like DNNs.
- C. **It ensures higher prediction accuracy than post-hoc interpretability.** – The opposite is true.
- D. **It provides global and local levels of interpretability.**

Pitfall: Mixing built-in interpretability with post-hoc interpretability

Question 4: In the study conducted by Adebayo et al., which ONE of the following was the surprising finding regarding some popular saliency attribution methods?

- A. **Superior AI Performance vs human performance** – This was not tested in the study.
- B. **CNN failure** – This generalization is incorrect.
- C. **Parameter Independence** - Some methods produced maps that didn't change even when model weights were randomized.
- D. **Normalisation** – This was irrelevant to what the study was mainly trying to find.

Pitfall: Assuming saliency maps are always model-dependent.

Question 5: In SmoothGrad, which specific parameter handles the intensity of the Gaussian noise?

- A. **M_c** – the resulting saliency map
- B. **σ** – The noise level σ sets the perturbation scale, with 10%-20% defined as the ideal range.
- C. **n** – the number of samples taken
- D. **σ^2** – The variance which is used in the formula but not to directly tune the parameters

Pitfall: Confusing variance notation with the actual parameter used in implementation.

Question 6: Which ONE of the following is the reason why a 'vanilla' gradient is unreliable when a feature 'saturates'?

- A. Vanishing local derivatives** - *If the function flattens out, the gradient approaches zero even if the feature is important.*
- B. Exploding attribution values** – *The opposite of what happens during saturation*
- C. Negative pixel handling** – *This is unrelated to saturation*
- D. Random noise** – *Saturation causes the gradient to approach zero, not randomness*

Pitfall: Assuming gradients always reflect importance.

Question 7: By which ONE of the following means does Guided Backpropagation modify the backward pass?

- A. ReLU gradient clipping** – *It blocks negative gradients to focus only on the features that help the neuron fire.*
- B. Average noisy inputs** – *This option describes SmoothGrad*
- C. Input multiplication** – *This refers to a different technique (Input x Gradient.)*
- D. Removing ReLU activations** – *This would break the model's structure.*

Pitfall: Confusing gradient filtering with gradient averages.

Question 8: Which ONE of the following requires saliency to stay consistent if the input is shifted by a constant?

- A. Input Invariance** – *Pre-processing that does not change the model's output should not fundamentally change the attribution.*
- B. Sensitivity** – *Relies on non-zero attribution for changes*
- C. Implementation Invariance** – *Compares two different models*
- D. Completeness** – *Relates to the attribution totals*

Pitfall: Mixing invariance with unrelated mathematical properties.

Question 9: Which ONE of the following sometimes causes saliency maps to show edges on a randomized model?

- A. Model-independent behaviour** – *Some methods act like edge detectors rather than showing learned features.*
- B. Output-only computation** - *Gradients are back-propagated, not just at the output.*
- C. Pixel ignoring** - *Models must process pixels to generate gradients.*
- D. Pre-trained feature extractors** – *These issues occur even without pre-training.*

Pitfall: Assuming visual quality implies meaningful explanation.

Question 10: As the sample size increases, which ONE of the following happens when the SmoothGrad technique is applied?

- A. Noise is increased.** – *The opposite happens. More samples decrease noise through averaging.*
- B. The saliency map vanishes to black.** – *This saliency map always remains visible because SmoothGrad averages the gradients of the saliency maps. SmoothGrad ensures that features are still visible even if some gradients are zero.*
- C. Sharpness is affected.** – *Sharpness is a result of averaging, not just scaling σ .*
- D. Returns diminish after $n > 50$.** – *As more trials are averaged, maps get clearer, but the benefit levels off.*

Pitfall: Assuming "more noise inputs = noisier output" instead of averaging effect.